APPENDIX I

Orbit theory

I.1. Space story

The author's wife came to watch one of the GSTAR III orbit-raising maneuvers. We did over 160 maneuvers over the course of the recovery. Each maneuver meant turning on the Electrothermal Hydrazine Thrusters (EHTs). The maneuver involved closing the pitch loop, heating up the thrusters and turning them on for the burn duration. We then shut everything down returning one orbit period later. I asked her what she thought. She said, "It was boring."

I.2. Introduction

This chapter discusses orbit representation and dynamics. It provides a brief introduction to orbit geometry followed by a discussion of orbit representations. Many spacecraft attitude disturbances are functions of the orbital position. For Earth- or planet-pointing spacecraft, such as low-Earth orbit constellations, it is necessary to know the orbital position to determine the pointing direction. Formation flying is another area of interest and in formation flying relative orbital motion is tied to the individual satellite pointing. In some cases the attitude control engineer can work with a precomputed orbital track. In other cases, it is necessary to numerically integrate the orbit along with the attitude.

I.3. Representations of orbits

I.3.1 Orbital geometry

An orbit involves at least two bodies, for example a planet and a spacecraft. In the ideal two-body case the two bodies rotate about the common center-of-mass, known as the barycenter. For all practical spacecraft cases, the spacecraft mass itself is negligible and this means that the satellite follows a conic section path about the primary body's center-of-mass. The orbit may be elliptical, with an eccentricity less than one, parabolic with an eccentricity equal to one or hyperbolic with an eccentricity greater than one. Elliptical orbits stay around the orbit center. Parabolic and hyperbolic orbits go to infinity. Fig. I.1 shows the geometry of an elliptical orbit. This is a planar orbit in which the orbital motion is two-dimensional.

The semimajor axis a is

$$a = \frac{r_a + r_p}{2} \tag{I.1}$$

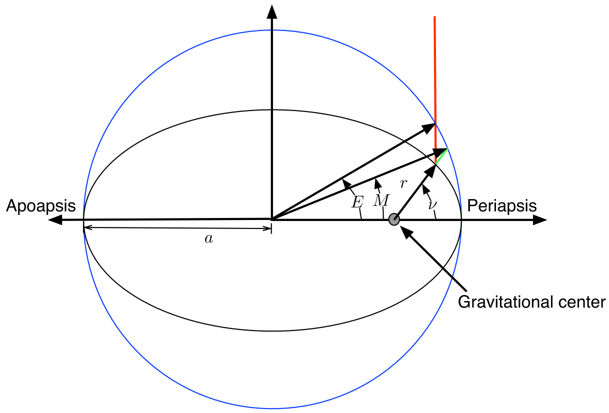


Figure I.1 Elliptical orbit.

where r_a is the apoapsis (apogee for Earth) radius, or point furthest from the central planet, and r_p is the periapsis radius (perigee for the Earth), or closest point to the planet. The eccentricity, e, of the orbit is

$$e = \frac{\frac{r_a}{r_p} - 1}{\frac{r_a}{r_p} + 1} \tag{I.2}$$

When $r_a = r_p$ the orbit is circular and $e = 0$. This formula is not meaningful for parabolic or hyperbolic orbits. Fig. I.1 shows three angular measurements, M is the mean anomaly, E is the eccentric anomaly, and ν is the true anomaly. All are measured from periapsis. The mean anomaly is related to the mean orbit rate n through a linear function of time and orbit rate

$$M = M_0 + n(t - t_0) \tag{I.3}$$

where n is the orbit rate and t_0 is the time when $M = M_0$. The eccentric anomaly is the angle to the current position as projected onto the ellipse's circumscribing circle, drawn in blue. It is related to the mean anomaly by Kepler's equation

$$M = E - e\sin E \tag{I.4}$$

where e is the orbit eccentricity. This equation needs to be solved numerically in general, but for small values of e this approximation can be used:

$$E \approx M + e\sin M + \frac{1}{2}e^2 \sin 2M \tag{I.5}$$

Higher-order approximations can also be found. The true anomaly is related to the eccentric anomaly through the equation

$$\tan\frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{E}{2} \tag{I.6}$$

where ν is the true anomaly. Finally, the orbit radius is

$$r = \frac{a(1-e^2)}{1+e\cos\nu} \tag{I.7}$$

If $e > 1$ in this equation r will go to ∞, as is expected for hyperbolic orbits. For a hyperbolic orbit, a is negative.

I.3.2 Cartesian coordinates

The central-body equations in Cartesian coordinates are

$$\ddot{r} + \mu\frac{r}{|r|^3} = a \tag{I.8}$$

where r is the position vector from the mass center of the central body, a is the acceleration vector of all other disturbances (such as thrusters, solar pressure, and gravitational accelerations due to other planets and planetary asymmetries), and μ is the gravitational parameter for the central body. The radius vector is

$$r = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \tag{I.9}$$

and the radius magnitude is $|r| = \sqrt{x^2+y^2+z^2}$. The equations are nonlinear. However, when the disturbance acceleration a is zero, r traces ellipses, parabolas, or hyperbolas. Equations can also be written in cylindrical or spherical coordinates. a also includes accelerations due to harmonics of the planet's gravity and other nonideal gravitational effects.

I.3.3 Keplerian elements

Seven parameters are necessary to define an orbit of a spacecraft about a spherically symmetric body. One is the gravitational parameter, generally denoted by the symbol μ. There are many ways of representing the other six elements. The most popular are position and velocity (r and v) vectors and Keplerian elements, both are shown in Fig. I.2.

The Keplerian elements are defined as follows: Two elements define the elliptical orbit. The size of the orbit is determined by the semimajor axis a, which is the average of the perigee radius and apogee radius. The shape of the orbit is defined by the

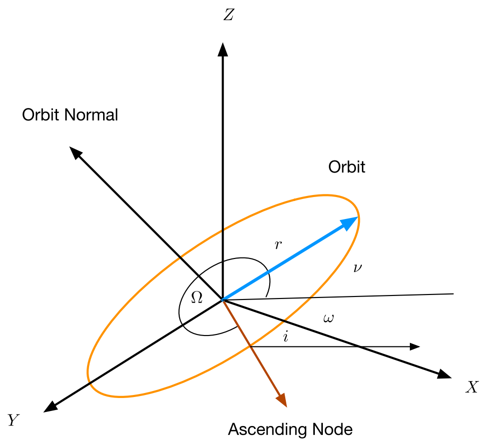


Figure I.2 Orbital elements.

eccentricity, e. Two elements define the orbital plane. Ω is the longitude or right ascension of the ascending node, or the angle from the $+X$-axis of the reference frame to the line where the orbit plane intersects the xy-plane. i is the inclination and is the angle between the xy-plane and the orbit plane. ω is the argument of perigee and is the angle in the orbit plane between the ascending node line and perigee (where the orbit is closest to the center of the central body). ν is the true anomaly and is the angle between perigee and the spacecraft. The mean anomaly M may be used in the element set instead of ν. M or ν tell us where the spacecraft is in its orbit. To summarize, the Keplerian elements are

$$x = \begin{bmatrix} a \\ i \\ \Omega \\ \omega \\ e \\ M \end{bmatrix} \tag{I.10}$$

The variational equations for Keplerian elements are

$$p = a(1 - e^2) \tag{I.11}$$

$$r = \frac{p}{1 + e\cos\nu} \tag{I.12}$$

$$h = \sqrt{\mu p} \tag{I.13}$$

$$b = \sqrt{pa} \tag{I.14}$$

$$n = \frac{h}{ab} \tag{I.15}$$

where h is angular momentum, p is the parameter, i is the inclination, ν is the true anomaly, M is the mean anomal, and n is the orbital rate. The accelerations are a_R, radial, a_T, tangential, and a_N normal to the orbit.

$$\dot{a} = \frac{2a^2(e \sin \nu a_R + \frac{p}{r} a_T)}{h} \tag{I.16}$$

$$\dot{i} = \frac{r \cos(\omega + \nu) a_N}{h} \tag{I.17}$$

$$\dot{\Omega} = \frac{r \sin(\omega + \nu) a_N}{h \sin i} \tag{I.18}$$

$$\dot{\omega} = \frac{\frac{-p \cos \nu a_R + (p+r) \sin \nu a_T}{e} - \frac{r \sin(\omega+\nu) \cos i a_N}{\sin i}}{h} \tag{I.19}$$

$$\dot{e} = \frac{p \sin \nu a_R + (p + r) \cos \nu + re) a_T}{h} \tag{I.20}$$

$$\dot{M} = n + b\frac{p(\cos \nu - 2re) a_R - (p + r) \sin \nu a_T}{ahe} \tag{I.21}$$

For near-circular orbits, $e = 0$ so $p = a$ and $r = a$. We are interested only in the first three equations that become

$$\dot{a} = \frac{2a^{3/2} a_T}{\sqrt{\mu}} \tag{I.22}$$

$$\dot{i} = \sqrt{\frac{a}{\mu}} \cos(\omega + \nu) a_N \tag{I.23}$$

$$\dot{\Omega} = \sqrt{\frac{a}{\mu}} \left(\frac{\sin(\omega + \nu)}{\sin i} \right) a_N \tag{I.24}$$

I.3.4 Equinoctial elements

Modified equinoctial coordinates are a form of orbital elements that are not singular when e is zero nor i is zero. The modified coordinates are

$$\begin{aligned} p &= a(1 - e^2) \\ f &= e \cos(\omega + \Omega) \\ g &= e \sin(\omega + \Omega) \\ h &= \tan(i/2) \cos \Omega \\ k &= \tan(i/2) \sin \Omega \\ L &= \Omega + \omega + \nu \end{aligned} \tag{I.25}$$

where p is the semiparameter, a is the semimajor axis, e is the orbital eccentricity, ω is the argument of perigee, Ω is the right ascension of the ascending node, L is the true longitude, and ν is the true anomaly.

The dimensional dynamical equations are

$$\dot{x} = Ga + b \tag{I.26}$$

where a is the acceleration vector with components in the radial, tangential, and normal directions and includes all accelerations except for point-mass gravity, and b drives the true longitude, L.

The state vector is

$$x = \begin{bmatrix} p \\ f \\ g \\ h \\ k \\ L \end{bmatrix} \tag{I.27}$$

and

$$G = \begin{bmatrix} 0 & 2r\omega & 0 \\ \omega s & \omega(\gamma c + f)/q & -\Omega g \\ -\omega c & \omega(\gamma s + g)/q & -\Omega f \\ 0 & 0 & \zeta c \\ 0 & 0 & \zeta s \\ 0 & 0 & \Omega \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \sqrt{\frac{\mu p}{r^2}} \end{bmatrix} \tag{I.28}$$

where

$$r = \frac{p}{q} \tag{I.29}$$

$$c = \cos L \tag{I.30}$$

$$s = \sin L \tag{I.31}$$

$$z = hs - kc \tag{I.32}$$

$$\omega = \sqrt{\frac{p}{\mu}} \tag{I.33}$$

$$q = 1 + fc + gs \tag{I.34}$$

$$\gamma = q + 1 \tag{I.35}$$

$$\Omega = \omega\frac{z}{q} \tag{I.36}$$

$$\zeta = \frac{1}{2}\frac{\omega}{q}(1 + \sqrt{h^2 + k^2}) \tag{I.37}$$

If the accelerations are zero, only L changes. The equinoctial acceleration frame shown in Fig. I.3 shows the definition of the acceleration vector a that multiplies G.

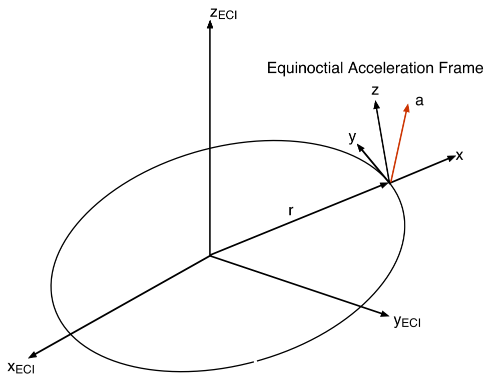


Figure I.3 Equinoctial acceleration frame.

We define the unit vectors as

$$\hat{x} = \frac{r}{|r|} \tag{I.38}$$

$$\hat{y} = \frac{n \times s}{|n \times s|} \tag{I.39}$$

$$\hat{z} = n \tag{I.40}$$

where

$$n = \frac{r \times v}{|r \times v|} \tag{I.41}$$

The transformation matrix from inertial to equinoctial is

$$C_{IToE} = \begin{bmatrix} \hat{x}^T \\ \hat{y}^T \\ \hat{z}^T \end{bmatrix} \tag{I.42}$$

I.4. Propagating orbits

I.4.1 Introduction

There are four general methods for propagating an orbit. One is Kepler propagation. For example, if there are no perturbing accelerations on the spacecraft (such as thruster firings, solar pressure, or the gravitational pull of other planets) it is possible to just propagate the elements directly. More complex propagators can account for some kinds of perturbations.

The second method is an analytical theory. This is used for planets and their moons. Very complex nonlinear expressions can be used to predict the position of a planet or moon with high accuracy over long periods of time. Some of these analytical theories are hundreds of pages long. While rarely used for attitude or orbit control work, they can be useful onboard flight computers if it is necessary to predict the position of an astronomical body.

The third method is numerical integration, which is used for most practical orbit modeling. It allows the engineer to incorporate all perturbations in a straightforward fashion and permits easy modification of the models for additional disturbance sources.

The fourth method uses what are known as two-line elements. These are sometimes called NORAD two-line elements. They are orbital fits that include perturbing accelerations like drag, solar pressure, and the higher harmonics of the Earth's gravity. Two-line elements are available for all orbiting spacecraft.

I.4.2 Kepler propagation

If perturbing accelerations are small, it is possible to propagate the orbital elements directly. Kepler propagation is very useful when you just need to test a control system as the spacecraft orbits the planet and you are not concerned with what is happening to the orbit as a result of thruster firings, etc. To determine the orbit it is necessary to know the eccentric anomaly. The equation that must be solved for obtaining the eccentric anomaly from elliptical orbits is

$$M = E - e\sin E \tag{I.43}$$

where e is eccentricity and E is the eccentric anomaly, which is known as Kepler's equation. It is necessary to solve for E iteratively but this can be done in a straightforward fashion. Impulse changes in velocity can be accommodated easily. To change the orbit with an impulse change in velocity in which the velocity changes while the position stays fixed just:

1. Convert the orbital elements to r and v vectors;
2. Add the Δv to v;
3. Convert $v + \Delta v$ and r back to orbital elements.

This procedure is useful for modeling solid-rocket burns or any Δv that happens in a time much less than the orbit period.

I.4.3 Numerical integration

The state equations for orbit propagation are

$$\dot{v} = -\mu \frac{r}{|r|^3} + g(r) + f(\theta) + h \tag{I.44}$$

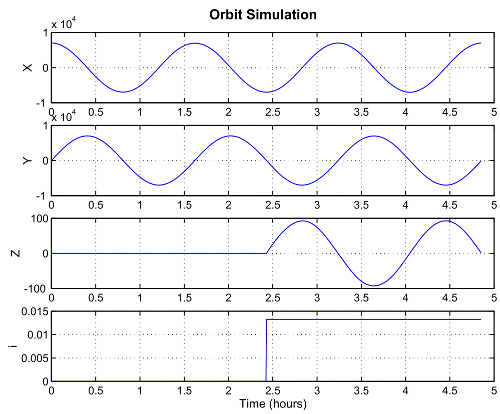
$$\dot{x} = v \tag{I.45}$$

The terms on the right-hand side of the velocity-derivative equation are the spherical gravity acceleration, additional accelerations g that are functions of position, accelerations f that are functions of angles, and accelerations h that are independent of orientation or position (such as thruster firings). Orientation-dependent forces include aerodynamic drag and solar-pressure forces. The orientation may just be the orientation of the spacecraft with respect to inertial space, or may include the relative orientation of components such as solar arrays. If the orientation forces are small, then the orbit equations may be decoupled from the attitude equations and propagated separately. Otherwise, they must be integrated together. For example, the drag on the Space Shuttle in low-Earth orbit is dependent on orientation so the attitude and orbit equations cannot be decoupled.

For precise orbit predictions it is necessary to use integrators with error correction. The most widely used today are the Runge–Kutta n/n+1-order integrators in which coefficients of the equations are chosen so that with minimal extra computation a nth-order and $n+1$th-order solution are computed. The results can be compared and the step size adjusted. Some integrators will also change order to minimize the solution error.

The following example shows an orbit simulation which uses a fourth-order Runge–Kutta integrator with the Cartesian orbit equations. Example I.1 shows a simulation of a circular orbit. In this case we suddenly change the z velocity from 0 to 0.1 km/s half-way through the simulation. This causes an instantaneous change in inclination.

```
nSim         = 1000;
xPlot        = zeros(4,nSim);
[r,v]        = El2RV( [7000 0 0 0 0 0 ] );
tEnd         = 3*Period(7000);
dT           = tEnd/nSim;
x            = [r;v];
for k = 1:nSim
  el           = RV2El( x(1:3), x(4:6) );
  xPlot(:,k) = [x(1:3);el(2)];
  x            = RK4( 'FOrbCart', x, dT, 0 );
  if( k == 500 )
    x(6) = 0.1;
  end
end
[t,c] = TimeLabl( (0:(nSim-1))*dT );
Plot2D( t, xPlot, c, ['X';'Y';'Z';'i'], 'Orbit Simulation' )
PrintFig(1,1,1,'OrbitSim')
```



Example I.1: Orbit simulation of impulsive inclination change.

El2RV converts elements in an array $[a, i, w, W, e, M]$ to r and v. Period returns the period of the orbit given the semimajor axis. RV2El computes r and v from the orbital elements. RK4 is the 4th-order Runge–Kutta integrator. It does not perform step-size adjustment. When $k = 500$ we instantaneously change the z-velocity. This is a good approximation to an impulsive velocity change, which you might get with a solid rocket motor firing.

x, y, and z all look like harmonic oscillators, as shown above. The inclination change is evident in the plot. Many types of mission planning can be done as simply as the example shown here.

I.5. Gravitational acceleration

I.5.1 Point masses

The simplest possible model for the gravitational acceleration due to a planet is to assume that the planet is a point mass. That means it has no dimensions but finite mass. In this case the acceleration due to a planet on a second body is

$$a = -\mu \frac{r}{|r|^3} \tag{I.46}$$

where r is the vector from the barycenter.

The planet has finite dimensions and since orbits are always outside the planet, the singularity at $|r| = 0$ is not important. The above equation is written with respect to a coordinate frame that is inertially fixed. Since the gravitational acceleration depends only on the radius the latitude and longitude of the subsatellite point (the point on the r vector that is on the planet's surface) is not important. This will not be true when planetary asymmetries are accounted for later.

Assume that there are n gravitational sources, all point masses, then the equations of motion for the satellite can be written as follows.

$$\begin{aligned} a &= -\mu \frac{r}{|r|^3} - \mathrm{G}\sum_{j=3}^{N} m_j \left(\frac{d_j}{|d_j|^3} + \frac{\rho_j}{|\rho_j|^3} \right) \qquad \text{(I.47)} \\ \mu &= \mathrm{G}(m_1 + m_2) \end{aligned}$$

where the variables are described in Fig. I.4. The primary planet is the object that produces the largest gravitational acceleration on the spacecraft. It is body 1 and the spacecraft is body 2.

The above equations are important for all Earth-orbiting satellites. For Earth satellites the Sun and Moon generate significant perturbations in the orbit and must be included. For lunar orbits, the Earth and the Sun should be included.

I.5.2 Planetary asymmetries

There are many ways to deal with asymmetries in planets. One would be to model the planet as a set of point masses contained within the surface of the planet. The second

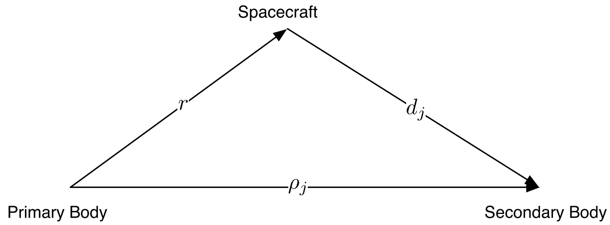


Figure I.4 Position vectors in the *n*-body system.

is to expand the gravitational potential in a spherical-harmonic expansion and compute the gravity forces from the gradient of the potential. Other expansions could also be used.

The spherical-harmonic expansion method works as follows. Define the perturbing gravitational potential of a planet as

$$V = \frac{\mu}{r}\sum_{n=2}^{\infty}\left[\left(\frac{a}{r}\right)^n \sum_{m=0}^{\infty}\big(S_{n,m}\sin m\lambda + C_{n,m}\cos m\lambda\big)\, P_{n,m}(\sin\phi)\right] \tag{I.48}$$

defined in the planet fixed frame. a is the radius of the planet. The definition of the coordinates is shown in Fig. I.5.

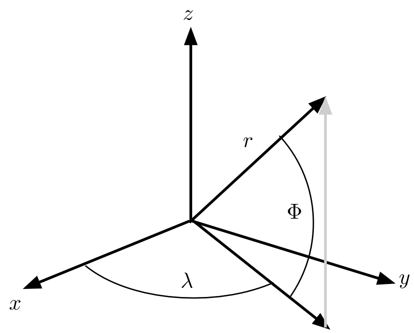


Figure I.5 Coordinates for a nonhomogeneous planet.

If we define i, j, k as unit vectors along the planetary x-, y-, and z-axes, the gravitational acceleration can be found as follows

$$\frac{\partial V}{\partial r} = \sum_{n=2}^{\infty}\sum_{m=0}^{n}\frac{\partial V_{n,m}}{\partial r} \tag{I.49}$$

and

$$\frac{\partial V_{n,m}}{\partial r} = \frac{\mu}{r^2}\left(\frac{a}{r}\right)^n \left[-r(\nu H_{n,m} + B_{n,m}) + iD_{n,m} - jE_{n,m} + kH_{n,m}\right] \tag{I.50}$$

$$B_{n,m} = (C_{n,m}\hat{C}_m + S_{n,m}\hat{S}_m)(n+m+1)P_n^m \tag{I.51}$$

$$E_{n,m} = -m(C_{n,m}\hat{S}_{m-1} - S_{n,m}\hat{C}_{m-1})$$

$$D_{n,m} = m(C_{n,m}\hat{C}_{m-1} + S_{n,m}\hat{S}_{m-1})$$

$$H_{n,m} = (C_{n,m}\hat{C}_m + S_{n,m}\hat{S}_m)P_n^{m+1}$$

$$\hat{C}_m = \hat{C}_1\hat{C}_{m-1} - \hat{S}_1\hat{S}_{m-1}$$

$$\hat{S}_m = \hat{S}_1\hat{C}_{m-1} + \hat{C}_1\hat{S}_{m-1}$$

The starting conditions are

$$\begin{array}{ll} \hat{C}_0 = 1 & \hat{S}_0 = 0 \\ \hat{C}_1 = \frac{x}{r} & \hat{S}_1 = \frac{y}{r} \end{array} \tag{I.52}$$

The derivatives of the Legendre polynomials are

$$P_n^0 = \frac{1}{n}\left[(2n-1)\nu P_{n-1}^0 - (n-1)P_{n-2}^0\right] \tag{I.53}$$

$$P_n^m = P_{n-2}^m + (2n-1)P_{n-1}^{m-1} \tag{I.54}$$

$$P_0^0 = 1$$

$$P_1^0 = \frac{z}{r} \equiv \nu$$

$$P_0^1 = 0$$

$$P_1^1 = 1$$

This kind of harmonic expansion is the standard method for modeling a planet's gravitational field. Harmonic models are available for the Earth, Moon, Mars, and other planets. A harmonic model is convenient because it gives an idea of how much influence higher-order terms have on the overall force.

An alternative approach is to use (I.47) and model the potential using a set of point masses. A combination of point masses and a spherical harmonic model can also be employed. This has been used for lunar-gravity models where concentrations of mass (mass cons) are modeled by point masses and the overall figure of the Moon is modeled using a harmonic expansion.

Sometimes the harmonics with $m = 0$ are given as J_n starting with J_2 ($J_1 = 0$). The largest term for the Earth is J_2, which is -1.0826×10^{-3}, due to the oblateness (or flatness of the poles compared to the equator) of the Earth. J_3 is 2.5326×10^{-6}. J_n is the same as $C_{n,0}$.

I.6. Linearized orbit equations

The central-body equations in cylindrical coordinates are

$$\ddot{r} - r\dot{\theta}^2 + \frac{\mu}{r^2} = a_x \cos\theta + a_y \sin\theta \tag{I.55}$$

$$r\ddot{\theta} + 2\dot{r}\dot{\theta} = a_y \cos\theta - a_x \sin\theta \tag{I.56}$$

$$\ddot{z} + \frac{\mu z}{r^3} = a_z$$

$$r^2 = x^2 + y^2 + z^2$$

These equations are still nonlinear. The linearized equations for a given R and constant orbit rate ω are

$$\ddot{x} - 3\omega^2 x - 2\omega\dot{y} = a_x \tag{I.57}$$

$$\ddot{y} + 2\omega\dot{x} = a_y \tag{I.58}$$

$$\ddot{z} + \omega^2 z = a_z \tag{I.59}$$

where x is radial deviation from the nominal R, y is $R\theta$ where θ is the orbit angle to the reference and R is the nominal orbit radius, and z is normal to the orbit. The disturbances have been redefined. Define $\sigma = \omega t$. The equations become

$$\ddot{x} - 3x - 2\dot{y} = \frac{a_x}{\omega^2} \tag{I.60}$$

$$\ddot{y} + 2\dot{x} = a_y \tag{I.61}$$

$$\ddot{z} + z = a_z \tag{I.62}$$

where the dot is now with respect to σ. These new equations are much better suited for numerical analysis than the previous set. This is because the resulting elements in the state-transition matrix are all the same order of magnitude. When the elements vary greatly in magnitude the smaller ones tend to get lost in many numerical routines. An interesting problem is that of a solar sail in orbit around the Sun with the sail normal along the radius vector. This results in a purely radial acceleration. The equations become

$$\ddot{x} - 3x - 2\dot{y} = \frac{a_x}{\omega^2} \tag{I.63}$$

$$\ddot{y} + 2\dot{x} = 0$$

The easiest way to solve this problem is with Laplace transforms,

$$\begin{bmatrix} s^2 - 3 & -2s \\ 2s & s^2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{a_x}{s\omega^2} \\ 0 \end{bmatrix} \tag{I.64}$$

or

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{\begin{bmatrix} s^2 & 2s \\ -2s & s^2-3 \end{bmatrix}}{s^2(s^2+1)} \begin{bmatrix} \frac{a_x}{s\omega^2} \\ 0 \end{bmatrix} \tag{I.65}$$

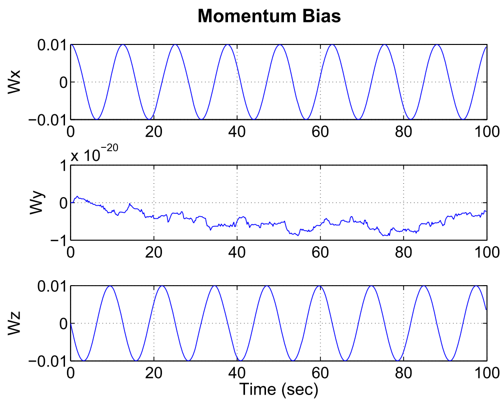
The solution is

$$\begin{aligned} x &= a_x(1-\cos\omega t) \\ y &= -2a_x(\omega t - \sin\omega t) \end{aligned} \tag{I.66}$$

A radial force causes an oscillation in the radial direction, but also causes the spacecraft to move in the circumferential direction relative to the initial position. Thus a radial force can be used to change the phase of the spacecraft.

The response of the inplane components of a linearized orbit is shown in Example I.2. The orbit rate is set to 1.

```
[a,b]    = LinOrb( [], 1 );
a        = a([1 2 4 5],[1 2 4 5]);
b        = b([1 2 4 5],[1,2] );
c        = eye(2,4);
d        = eye(2,2);
[m,p,w]  = FResp(a,b,c,d);
yL       = {'|1,1|', '|1,2|', '|2,1|', '|2,2|'};
Plot2D(w,20*log10(m),'\omega (rad/s)', yL,...
  'Orbit Frequency response', 'xlog')
```



Example I.2: Linearized orbit-frequency response.

The z component is a simple oscillator. The crosschannels (i.e., disturbance in 2 to 1 position) have only zeros, no poles. The diagonal channels have pole zero pairs. All of the poles are at orbit rate. A disturbance in the radial direction will cause a steady offset in radius. However, a step radial disturbance will cause an infinite steady response in tangential position and a step tangential disturbance will cause an infinite steady response in body axes.